

Module: **Problem 4**

This module provides a utility function for determining whether a given integer (or integer-compatible value) is a triangular number. A triangular number T_n can be expressed as $n(n+1)/2$ for some positive integer n . The implementation uses the inverse triangular-number formula and checks if the resulting index is an integer.

Functions

`is_triangular(k)`

Description:

Determines whether the argument `k` represents a triangular number. The function computes

$$[x = \frac{\sqrt{8k + 1} - 1}{2}]$$

and returns `True` if `x` is an integer (i.e., `k` fits the triangular-number formula), otherwise `False`.

Parameters: - `k` (*int or float or bool*): The value to test. The docstring states it should be a positive integer, but the implementation accepts any numeric type that can be used in arithmetic with `math.sqrt`. Non-numeric arguments will raise a `TypeError`.

Returns:

`bool` - `True` if `k` is triangular, `False` otherwise.

Examples:

```
# Based on observed runtime behavior
result = is_triangular(0)      # → True  (0 is treated as triangular by this implementation)
result = is_triangular(1)      # → True
result = is_triangular(True)   # → True  (True is interpreted as 1)
```

Edge Cases / Notes: - The function does **not** validate that `k` is positive. Supplying a negative integer (e.g., `-1`) triggers `math.sqrt` on a negative argument, raising `ValueError: math domain error`. - Passing non-numeric types (e.g., strings) causes a `TypeError` because the expression `8*k + 1` attempts to concatenate a string with an integer. - `bool` values are subclasses of `int`; therefore `True` is treated as `1` and returns `True`. - Mathematically, the triangular sequence starts at `1` ($T_1 = 1$). The current implementation incorrectly returns `True` for `k = 0`. Users may need to add an explicit check if `0` should be excluded.